

$$PQ = \begin{pmatrix} 1 & 2 & \cdots & n \\ p_1 & p_2 & \cdots & p_n \end{pmatrix} \begin{pmatrix} 1 & 2 & \cdots & n \\ q_1 & q_2 & \cdots & q_n \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & \cdots & n \\ p_{q_1} & p_{q_2} & \cdots & p_{q_n} \end{pmatrix}$$

$$(12)(13) = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$= (132)$$

Multiplication table of S_3

S_3	e	(132)	(123)	(23)	(13)	(12)
e	e	(132)	(123)	(23)	(13)	(12)
(132)	(132)	(123)	e	(12)	(12)	(23)
(123)	(123)	e	(132)	(13)	(23)	(13)
(23)	(23)	(12)	(13)	e	(123)	(132)
(13)	(13)	(23)	(12)	(132)	e	(123)
(12)	(12)	(13)	(23)	(123)	(132)	e

$$\begin{array}{rcl} & 1 & \longrightarrow 3 \longrightarrow 2 \\ (23)(132) & \longrightarrow & 2 \longrightarrow 1 \longrightarrow 1 \\ & 3 & \longrightarrow 2 \longrightarrow 3 \end{array}$$

The dihedral group

$$D_n \longrightarrow \begin{array}{l} \bullet \text{ Rotations } \text{ angle } \frac{2\pi}{n} \\ \bullet \text{ Reflection} \end{array}$$

presentation

$$D_n = \langle r, \sigma \mid r^n = e, \sigma^2 = e, \sigma r \sigma = r^{-1} \rangle \quad |D_n| = 2n$$

$$D_3 = \left\{ \underbrace{e, r, r^{-1}}_{\text{rotations}}, \underbrace{\sigma_1, \sigma_2, \sigma_3}_{\text{reflections}} \right\}$$

$$(\sigma_1 \sigma_2) \begin{array}{c} 1 \\ \triangle \\ 2 \quad 3 \end{array} = \sigma_1 \begin{array}{c} 3 \\ \triangle \\ 2 \quad 1 \end{array} = \begin{array}{c} 3 \\ \triangle \\ 1 \quad 2 \end{array} = r$$

Multiplication table of D_3

D_3	e	r	r^{-1}	σ_1	σ_2	σ_3
e	e	r	r^{-1}	σ_1	σ_2	σ_3
r	r	r^{-1}	e	σ_3	σ_1	σ_2
r^{-1}	r^{-1}	e	r	σ_2	σ_3	σ_1
σ_1	σ_1	σ_2	σ_3	e	r	r^{-1}
σ_2	σ_2	σ_3	σ_1	r^{-1}	e	r
σ_3	σ_3	σ_1	σ_2	r	r^{-1}	e

Generators and rank of a group

Def. of generators

A subset $S \subseteq G$ is called a set of generators of G if every element of G can be written as a finite product of elements of S and their inverses

$$\boxed{G = \langle S \rangle}$$

Ex Cyclic group C_3

$$C_3 = \{e, g, g^2\}, \quad g^3 = e$$

$$\langle g \rangle = C_3$$

$$\text{rank}(C_3) = 1$$

Ex Dihedral $D_3 = \{e, r, r^{-1}, \sigma_1, \sigma_2, \sigma_3\}$

$$\sigma = \sigma_1$$

$$\sigma_2 = r\sigma_1 = r\sigma$$

$$\sigma_3 = r^2\sigma = r^{-1}\sigma$$

$$D_3 = \langle r, \sigma \rangle \quad \text{rank}(D_3) = 2$$

Group morphism

Homomorphism

G (example, law $\circ$)

H (comp. law $\diamond$)

A homomorphism is a mapping

$$f^\circ : G \longrightarrow H, \quad g \longmapsto h = f(g)$$

Such that $g_1 \circ g_2 = g_3 \subset G$

$$\text{Then} \quad f(g_1) \diamond f(g_2) = f(g_3) \subset H$$

$$D_3 = \{e, r, r^{-1}, \sigma_1, \sigma_2, \sigma_3\} \quad C_2 = \{e, c\}, \quad c^2 = e$$

$$f^\circ : D_3 \longrightarrow C_2$$

$$e, r, r^{-1} \longrightarrow e \quad \sigma_1, \sigma_2, \sigma_3 \longrightarrow c$$

$$\textcircled{*} \quad \sigma_1 \circ \sigma_2 = r \quad \Longrightarrow \quad f(\sigma_1 \circ \sigma_2) = f(r) = e \subset C_2$$

$$f(\sigma_1) \diamond f(\sigma_2) = c \diamond c = e \subset C_2$$

$$\circledast \quad r \circ \sigma_1 = \sigma_2$$

$$f(r \circ \sigma_1) = f(\sigma_2) = c \quad \checkmark$$

$$f(r) \diamond f(\sigma_1) = e \diamond c = c$$

Counter example

$$\begin{cases} e \circ \sigma_3 \circ r^{-1} & \longrightarrow & e \\ \sigma_1 \circ \sigma_2 \circ r^{-1} & \longrightarrow & c \end{cases}$$

$$\sigma_3 \circ r^{-1} = \sigma_2$$

$$f(\sigma_3 \circ r^{-1}) = f(\sigma_2) = c \quad \times$$

$$f(\sigma_3) \diamond f(r^{-1}) = e \diamond e = e$$

Kernel

Kernel of a homomorphism f measures “how much” of the structure of G collapses on the identity of the map f .

$$\text{Ker}(D_3) = \{e, r, r^{-1}\}$$

Isomorphism

$$\longrightarrow \text{ bijective homomorphism } \quad G \cong H$$

For finite group isomorphic $\equiv$ identical multiplication table

$$C_4 = \{e, g, g^2, g^3\}, \quad g^4 = e$$

$$G_1 = \{1, i, -1, -i\}$$

$$G_3 = \{I, i\sigma_2, -I, -i\sigma_2\}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\implies \quad C_4 \cong G_1 \cong G_3$$

Endomorphism

$$\longrightarrow \text{ homomorphism of a group } G \text{ on itself } \quad f^\circ : G \longrightarrow G$$

$$\text{Ex: } f^\circ : \mathbb{R} \longrightarrow \mathbb{R}, \quad f(x) = 2x \quad (\text{with addition})$$

Automorphism

$$\longrightarrow \text{ bijective endomorphism}$$

The set of all automorphism of a group G is denoted

$$\text{Aut}(G) = \{f : G \longrightarrow G \mid f \text{ bijective and } f(gh) = f(g)f(h)\}$$

Ex group $(\mathbb{Z}, +) \longrightarrow$ two automorphism

$$f(n) = n \quad f(n) = -n$$

$$\text{Therefore} \quad \text{Aut}(\mathbb{Z}, +) \cong C_2$$

Cayley's Theorem

Every finite group of order n is isomorphic to a subgroup of the symmetric group S_n